Alice's Math Cheat Sheet

2026-11-02

Negative Exponents

$a^{-n} = \frac{1}{a^n}$ $\frac{1}{a^{-n}} = a^n$

Negative exponent? Move the base across the fraction bar and make the exponent +.

EXAMPLES

Rewrite: $7^{-2} = \frac{1}{7^2}$

Evaluate: $2^{-3} = \frac{1}{2^3} = \frac{1}{8}$

Evaluate: $\frac{1}{5^{-2}} = 5^2 = 25$

WATCH OUT

Only the base moves: $3y^{-4} = \frac{3}{y^4}$

It's not negative: $2^{-3} \neq -8$

Scientific Notation

$a \times 10^n$, where $1 \leq a < 10$

Number ≥ 10: positive exponent

Number < 1: negative exponent

Exponent = places the decimal moves

EXAMPLES

$45{,}000 = 4.5 \times 10^4$

$0.0072 = 7.2 \times 10^{-3}$

$3.2 \times 10^5 = 320{,}000$

Multiply: multiply numbers, add exponents

$(5 \times 10^3)(6 \times 10^4) = 30 \times 10^7$

$= 3 \times 10^8$

WATCH OUT

Number part ≥ 10? Move the decimal one place left and add 1 to the exponent.

Exponent Rules (same base)

Multiply: add $a^m \cdot a^n = a^{m+n}$

Divide: subtract $\frac{a^m}{a^n} = a^{m-n}$

Power of a power: multiply $(a^m)^n = a^{m \cdot n}$

No exponent shown? It's 1: $x = x^1$

EXAMPLES

$x^5 \cdot x^{-3} = x^{5+(-3)} = x^2$

$\frac{x^2}{x^{-4}} = x^{2-(-4)} = x^6$

$(x^{-2})^3 = x^{-6} = \frac{1}{x^6}$

Multiplying & Dividing Monomials

Numbers: multiply or divide them.

Exponents: add or subtract them.

EXAMPLES

$6m^4 \cdot (-2m^5) = -12m^9$

$-8k \cdot (-2k^3) = 16k^4$

$\frac{12x^7}{3x^2} = 4x^5$

WATCH OUT

Negative × negative = positive

Square & Cube Roots

$\sqrt{a}$: what number times itself makes a?

$\sqrt[3]{a}$: what number used 3 times makes a?

Perfect squares: 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144

Perfect cubes: 1, 8, 27, 64, 125, 216, 343, 512, 729, 1000

EXAMPLES

$\sqrt{49} = 7$ because $7 \cdot 7 = 49$

$\sqrt[3]{64} = 4$ because $4 \cdot 4 \cdot 4 = 64$

$\sqrt[3]{-8} = -2$ because $(-2)(-2)(-2) = -8$

$\sqrt{\frac{1}{9}} = \frac{1}{3}$ because $\frac{1}{3} \cdot \frac{1}{3} = \frac{1}{9}$

WATCH OUT

$\sqrt{25} = 5$, not 12.5 (don't divide by 2)

Multiplying Mixed Numbers

1. Change each to an improper fraction.
2. Multiply tops, multiply bottoms.
3. Simplify, then change back to mixed.

Improper: $2\frac{1}{3} = \frac{2 \cdot 3 + 1}{3} = \frac{7}{3}$

EXAMPLES

$2\frac{1}{3} \cdot 1\frac{1}{2} = \frac{7}{3} \cdot \frac{3}{2} = \frac{21}{6} = \frac{7}{2} = 3\frac{1}{2}$

WATCH OUT

Don't multiply wholes and fractions apart:

$2\frac{1}{3} \cdot 1\frac{1}{2} \neq 2\frac{1}{6}$

Dividing Mixed Numbers

1. Change each to an improper fraction.
2. Keep, Change, Flip: keep the first, change ÷ to ·, flip the second.
3. Multiply, simplify, change back to mixed.

Whole number? Put it over 1: $4 = \frac{4}{1}$

EXAMPLES

$3\frac{1}{2} \div 1\frac{3}{4} = \frac{7}{2} \div \frac{7}{4} = \frac{7}{2} \cdot \frac{4}{7} = \frac{28}{14} = 2$

$2\frac{2}{3} \div 4 = \frac{8}{3} \cdot \frac{1}{4} = \frac{8}{12} = \frac{2}{3}$

WATCH OUT

Flip only the second fraction, never the first.